Review report on
Fast and flexible inference for spatial extremes

Summary

The paper proposes *skewed Brown–Resnick* and *truncated extremal-t* processes whose exponent measure is continuous, enabling use of a computationally efficient spectral likelihood based on the associated intensity function. The authors describe inference for both max-stable and r-Pareto counterparts, and present simulations plus two applications (Florida rainfall and Red Sea SST) that suggest improved flexibility and speedups relative to composite likelihood and related methods.

While these ideas are promising, the manuscript in its present form suffers from (i) an emphasis on max-stable/r-Pareto models whose extremal dependence can be too rigid for environmental fields, (ii) untested modeling assumptions at the spatial scales considered, (iii) pervasive notational inconsistency that obscures key definitions and proofs, and (iv) incomplete evidence on computational efficiency at realistic data sizes. Substantial revision is therefore required.

Therefore, I recommend rejection of the paper in its current form; resubmission encouraged after major revisions

Major comments

(i) **Scope and title.** Recent work in spatial extremes has moved beyond asymptotic classes toward sub-asymptotic constructions such as random scale mixtures (e.g., Huser et al., 2017; Huser and Wadsworth, 2019) and max-infinitely-divisible models (e.g., Reich and Shaby, 2012; Bopp et al., 2021); see Huser and Wadsworth (2022) and Huser et al. (2024) for comprehensive reviews and comments.

In fact, Huser et al. (2024) cautioned the community to move away from max-stable process (or its corresponding r-Pareto process) for modeling environmental data due to their stringent dependence properties; see the following Major Comment (ii).

Given this context, the present title overstates generality. Please consider a more precise title such as *Fast spectral inference for max-stable and r-Pareto models*.

(ii) **The max-stability assumption**. The reason that the sub-asymptotic models are preferred over a max-stable process is not only because we want to get the dependence class right, but also because we want to use a model that can accommodate *weakening dependence* structure as the level of extremity increases so that the extreme clusters in the model will appear more localized as they become more intense. However, for any max-stable process

with unit Fréchet margins, the extremal coefficient of two locations with distance h becomes *almost constant* after a certain level of extremity:

$$\theta_h(z) - \theta_h = O(1/z), \text{ as } z \to \infty, \tag{1}$$

in which $\theta_h = 2 - \lim_{z\to\infty} \Pr\{Z(\boldsymbol{s}_1) > z | Z(\boldsymbol{s}_2) > z\}$. The corresponding r-Pareto also has this stability property. Although anisotropy in the proposed skewed Brown–Resnick may improve realism, property (1) remains and imposes a dangerous and unrealistic assumption on the storm size for extreme rainfall or heatwave extent for extreme temperature — spatial extremes at 100-year return level has almost the same size as storms at 200-year return level.

For real environmental datasets, I have rarely observed the extremal dependence leveling off as we focus on higher levels of extremity. This is also seen in Figure 5 of Huser and Wadsworth (2019). Therefore, max-stable process can be too rigid in many cases because of its stable dependence structure. Both Huser and Wadsworth (2019) and Hazra et al. (2025) can exhibit weakening dependence strength as the quantile level increases; see Figure 2 of Huser and Wadsworth (2019) for example.

Furthermore, the skewed Brown-Resnick process and the truncated extremal-t process is *asymptotically dependent everywhere* over a huge spatial domain considered in the applications of this paper; $\theta_h \to 2$ only if $h \to \infty$. This will *inflate the risks* associated with the simultaneous occurrences of the extremes over space. For example, the northern tip of the Red Sea ($\approx 29°56N, 32°34E$) and the southern tip ($\approx 12°29N, 43°20E$) are about 2,235 km apart in great-circle distance, for which asymptotic dependence is an absurd assumption.

(iii) **Model validity on large domains (need for testing).** Both proposed models are asymptotically dependent for any finite h on the large geographic domains analyzed. Before fitting, the manuscript should *test* max-stability/max-domain suitability, e.g., via the space-time max-stability test of Koh et al. (2025) (or comparable diagnostics). In my experience with SST fields (Hazra and Huser, 2021), max-stability is rejected even on subregions of 100 km, with rapid transition to asymptotic independence beyond modest lags. If a max-stable/r-Pareto model is used, restrict to subregions where the assumption is empirically defensible and say so explicitly.

It might be justifiable to use a max-stable or r-Pareto process to model the precipitation in a smaller spatial domain such as central Florida, but the authors need to specifically show that.

(iv) **Careless notations**. The authors use notations very frivolously, which causes immense confusion and makes the paper very difficult to read. For example, in Theorem 2, they wrote:

$$V(\boldsymbol{x}) = \sum_{k=1}^{D} \frac{1}{x_k} \Psi\left(\log\left(\frac{\boldsymbol{x}^{\circ\circ}_{-k}}{x^{\circ\circ}_k} \right) + \frac{\tilde{\boldsymbol{\omega}}^2_{-k} - \tilde{\omega}^2_k \mathbf{1}_{D-1}}{2} ; \boldsymbol{\mu}_k, \Sigma_k, \hat{\boldsymbol{\alpha}}_k, \tau_k \right), \tag{14}$$

where Ψ denotes the cdf of the extended skew-normal distribution (see Definition 1 in Section B of the Supplemental Material), $\boldsymbol{x}^{\circ\circ} = (x^{\circ\circ}_1, \ldots, x^{\circ\circ}_D)$, $x^{\circ\circ}_k = 2x_k\, \Phi(\tau_k; 0, 1)$, $k = 1, \ldots, D$, $\tau_k = \tilde{\omega}_k \delta_k$, $\tilde{\boldsymbol{\omega}} = (\tilde{\omega}_1, \ldots, \tilde{\omega}_D) = \omega \mathbf{1}$, $\omega =$

$\sqrt{\text{diag}(\Sigma)}$, $\boldsymbol{\delta} = \left(1 + \boldsymbol{\alpha}^\top \bar{\Sigma} \boldsymbol{\alpha}\right)^{-1/2} \bar{\Sigma} \boldsymbol{\alpha}$, $\bar{\Sigma} = \omega^{-1} \Sigma \, \omega$, $\boldsymbol{\mu}_k = A_k \Sigma_{\cdot k}$, $A_k = (\boldsymbol{e}_1, \ldots, \boldsymbol{e}_{k-1}, -\mathbf{1}, \boldsymbol{e}_k, \ldots, \boldsymbol{e}_{D-1})$, e_k, $k = 1, \ldots, D-1$, *are the standard basis vectors of dimension* $(D-1)$. *Moreover,* $\Sigma_k = A_k \, \bar{\Sigma} \, A_k^\top$, $\hat{\boldsymbol{\alpha}}_k = \left(1 - \boldsymbol{\delta}^\top \omega A_k^\top \Sigma_k^{-1} A_k \omega \boldsymbol{\delta}\right)^{-1/2} \hat{\omega}_k \, \Sigma_k^{-1} A_k \omega \boldsymbol{\delta}$, $\hat{\boldsymbol{\omega}}_k = \sqrt{\text{diag}(\Sigma_k)}$.

Excessive inline notations are very dense and hard to parse. Specifically,

- The bold (for vectors) and roman (for scalars) fonts are inconsistently used. For example, if $\tau_k = \tilde{\omega}_k \delta_k$ and $\tilde{\boldsymbol{\omega}} = (\tilde{\omega}_1, \ldots, \tilde{\omega}_D) = \omega \mathbf{1}$, why don't you simply write $\tau_k = \omega \delta_k$ (assuming ω is a scalar)?
  In $\bar{\Sigma} = \omega^{-1} \Sigma \, \omega$, are you multiplying two scalars to a matrix?
  In "$A_k = (\boldsymbol{e}_1, \ldots, \boldsymbol{e}_{k-1}, -\mathbf{1}, \boldsymbol{e}_k, \ldots, \boldsymbol{e}_{D-1})$, e_k, $k = 1, \ldots, D-1$, are …", what is e_k?
  How does "$\boldsymbol{\delta}^\top \omega A_k^\top \Sigma_k^{-1} A_k \omega \boldsymbol{\delta}$" and "$\hat{\omega}_k \, \Sigma_k^{-1} A_k \omega \boldsymbol{\delta}$" make any sense?
- The subscript in $\Sigma_{\cdot k}$ is not defined.
- The superscripts in $\frac{\boldsymbol{x}_{-k}^{\circ\circ}}{x_k^{\circ\circ}}$ are unnecessarily clumsy.
- The notations w_k and $\mathbb{I}$ in Equation (15) are not defined.
- The function Φ° in Equation (17) is never defined in the main text. I found its definition in (S.1) of Appedix B on Page 36!
- Under Equation (11), $\Pr(W \in [\mathbf{0}, \boldsymbol{x}r])$ also makes no sense.

There are many more notations that are not consistent throughout the text. The authors must carefully correct all the mistakes for this manuscript to be published in any outlet.

(v) **Computational evidence and baselines.** "More efficient than state-of-the-art" (Line 14 of Page 30) needs quantification. Please report wall-clock times and memory for simulation and both data sets, with/without anisotropy, knots, etc. and comparisons to contemporary baselines such as neural Bayes estimators (Sainsbury-Dale et al., 2025).
Moreover, the applications analyze $3,600$ (Florida rainfall) and $1,043$ (Red Sea SST) locations, whereas the simulation study uses only $D \in \{100, 225\}$; thus the simulation results do not demonstrate inferential performance at application scale.

(vi) Are 4–5 knots for $\boldsymbol{\eta}$ adequate on large, heterogeneous domains (Florida; Red Sea)? Related work uses richer local bases (e.g., 36 in Hazra et al., 2025 for a much smaller region). Please justify knot counts and provide sensitivity checks.

Minor comments

(i) Page 5 — If $C^+(S)$ is a function space and $B \subset C^+(S)$, how do you evaluate $\Pr(X/u \in B)$ if X is a real-valued spatial process?

(ii) Page 6 — You wrote "$\frac{(\tilde{W}(s))_+^\nu}{a(s)}$, where $\tilde{W}_+(s) = \max\{\tilde{W}(s), 0\}$". Please be consistent.

(iii) Equation (7) — Should "A" be "A_u"?

(iv) Page 8 — Should "$r(\boldsymbol{x}) = \|\boldsymbol{x}\|_1$" be "$r_0(\boldsymbol{x}) = \|\boldsymbol{x}\|_1$"?

(v) Page 15 — The constraint "$\boldsymbol{\eta}^\top \mathbf{1} = 0$ came out of no where. Why do you need it?

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